

6. Integration		
6.1	Evaluation of volume of revolution.	$\pi \int y^2 dx$ is required, but <i>not</i> $\pi \int x^2 dy$. Students should be able to find a volume of revolution, given parametric equations.
6.2	Simple cases of integration by substitution and integration by parts. Understand these methods as the reverse processes of the chain and product rules respectively.	Students will be expected to use a substitution to find, e.g. $\int x\sqrt{x-2} dx$ The substitution will be given in more complicated integrals. The integral $\int \ln x dx$ is required. More than one application of integration by parts may be required, for example, $\int x^2 e^x dx$, $\int e^x \sin x dx$.
6.3	Simple cases of integration using partial fractions.	Integration of rational expressions such as those arising from partial fractions, e.g. $\frac{2}{3x+5}$, $\frac{3}{(x-1)^2}$. Note that the integration of other rational expressions, such as $\frac{x}{x^2+5}$ and $\frac{2}{(2x-1)^4}$ is also required <i>(see P3 section 5.2).</i>
6.4	Analytical solution of simple first order differential equations with separable variables.	General and particular solutions will be required.
6.5	Use integration to find the area under a curve given its parametric equations.	Students should be able to find the area under a curve given its parametric equations. Students will not be expected to sketch a curve from its parametric equations.